

Fig. 3: Packaging detail inside a mobile handset with antenna

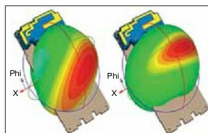


Fig. 4: GSM antenna radiation pattern for a cellphone

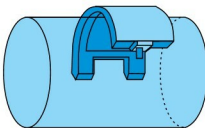


Fig. 5: Flexible microstrip applicator for hyperthermia medicinal applications

from which they are constructed or their geometrical shapes.

Microstrip antenna: the most common printed antenna

Microstrip antenna is one of the most popular types of printed antenna. It plays a very significant role in today's world of wireless communication systems. Microstrip antennae are very simple in construction using a conventional microstrip fabrication technique. Microstrip patch antenna consists of a radiating patch on one side of a dielectric substrate (FR4) that has a ground plane (Cu) on the other side as shown in Fig. 1.

The patch is generally made up of a conducting material such as copper or gold and can take any possible shape like rec-

tangular, circular, triangular, elliptical or some other common shape. The radiating patch and the feed lines are usually photo-etched on the dielectric substrate.

Microstrip patch antennae radiate primarily because of the fringing

fields between the patch edge and the ground plane. For good antenna performance, a thick dielectric substrate having a low dielectric constant (<6) is desirable since it provides higher efficiency, larger bandwidth and better radiation. However, such a configuration leads to a larger antenna size.

In order to design a compact microstrip patch antenna, a substrate with a higher dielectric constant (<12) must be used, which results in lower efficiency and narrower bandwidth. Hence a compromise must be reached between antenna dimensions and antenna performance. Excitation guides the electromagnetic energy source to the patch, generating negative charges around the feed point and positive charges on the other part of the patch. This difference in charges creates electric fields in the antenna that are responsible for radiations from the patch antenna.

Three types of electromagnetic waves are radiated. The first part is radiated into space, which is 'useful' radiation. The second part is diffracted waves, which are reflected back into space between the patch and the ground plane, contributing to the actual power transmission. The last part of the wave remains trapped in the dielectric substrate due to total reflection

TABLE I
Operating Frequency and Applications of Different Antennae

Antenna type	Frequency of operation	Application area
Wire antenna	300-3000 kHz (MF)	AM broadcasting, maritime radio, coast guard communication, direction finding
	3-30 MHz (HF)	Telephone, telegraph and facsimile, amateur radio, ship-to-coast and ship-to-aircraft communication
Aperture antenna	3-30 GHz (SHF)	Airborne radar, microwave links, satellite communication
Printed antenna	3-30 GHz (SHF)	Airborne radar, microwave links, satellite communication
	30-300 GHz (EHF)	Radar
Array antenna	300-3000 kHz (MF)	AM broadcasting, maritime radio, coast guard communication, direction finding
	3-30 MHz (HF)	Telephone, telegraph and facsimile, amateur radio, ship-to-coast and ship-to-aircraft communication
Reflector antenna	30-300 MHz (VHF)	Television, FM broadcast, air traffic control, police, navigational aids
	300-3000 MHz (UHF)	Television, satellite communication, surveillance radar, navigational aids
Lens antenna	30-300 MHz (VHF)	Television, FM broadcast, air traffic control, police, navigational aids
	300-3000 MHz (UHF)	Television, satellite communication, surveillance radar, navigational aids

Note: MF-medium frequency, HF-high frequency, VHF-very high frequency, UHF-ultra-high frequency, SHF-super-high frequency, EHF-extremely high frequency